Chinese Zero Pronoun Resolution: An Unsupervised Approach Combining Ranking and Integer Linear Programming

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Task: Zero Pronoun Resolution

Find an antecedent for an anaphoric zero pronoun (**AZP**).

What is a Zero Pronoun?

A zero pronoun (**zp**) is a gap in a sentence that is found when a phonetically null form is used to refer to a real-world entity.

Zero Pronoun Resolution: An Example

俄罗斯作为米洛舍夫维奇一贯的支持者，***pro***曾经提出调停这场政治危机。

Russia is a consistent supporter of Milošević, ***pro*** has proposed to mediate this political crisis.

The zero pronoun ***pro*** is coreferent with **俄罗斯(Russia)**.

Goal

Propose an unsupervised approach to this task

- Eliminate reliance on labeled data
  - Facilitate applicability to resource-scarce languages
- State-of-the-art approaches are supervised

Idea

Recast unsupervised ZP resolution as supervised ranking:

- Train a ranker in a supervised manner to resolve overt pronouns
- Apply the resulting ranker to resolve zero pronouns

Training the Ranker

Goal: Rank the candidate antecedents of an anaphoric overt pronoun so that its correct antecedents are ranked higher

- **Learning algorithm**
  - Yasmet, a maximum entropy-based ranking toolkit
- **Training instance creation**
  - Each instance represents an anaphoric overt pronoun ***op*** and one of its candidate antecedents ***c***
  - The set of instances created from the same anaphoric overt pronoun constitutes a ranking problem

$$rank\ value = \begin{cases} \dfrac{1}{|coref|}, & \text{if } op \text{ is coreferent with } c \\ 0, & \text{otherwise} \end{cases}$$

where |coref| is the number of correct antecedents ***op*** has

- **Feature set**
  - 32 features from state-of-the-art **AZP** resolvers (Zhao and Ng 2007; Chen and Ng 2013), including 4 features enforcing the compatibility between ***op*** and ***c*** w.r.t. 4 grammatical attributes, **Number**, **Gender**, **Person**, and **Animacy**

Applying the Ranker

Idea: Fill the gap of each anaphoric zero pronoun ***zp*** in the test set with each possible overt pronoun ***op***

- **Test instance creation**
  - For each anaphoric ***zp***, create a set of test instances from ***zp*** in the same way as the training instances
    - test instances are created for all possible combinations of ***zp***'s candidate antecedents ***c*** and gap-filling overt pronouns ***op*** except for those combinations where ***op*** and ***c*** are not compatible w.r.t. all of the 4 grammatical attributes
- **Antecedent selection**
  - Rank the test instances created from ***zp*** based on the probabilities *P*(***op***, ***c***) assigned to them by the ranker
  - Select as ***zp***'s antecedent the ***c*** in the instance having the highest probability

Using ILP to Improve Ranker's Output

Motivation: We enforced compatibility between ***op*** and ***c*** in a test instance w.r.t. the four grammatical attributes, but we didn't enforce such compatibility between ***op*** and its the governing verb

- Enforce such compatibility using Integer Linear Programming (ILP):

1. For each verb ***v*** in the test set, collect from the Chinese Gigaword corpus the set of NPs serving as the subject of ***v*** and then heuristically determine each NP's 4 grammatical attribute values
2. For each ***v***, computer its grammatical preferences, i.e., $P_a(b)$, the probability that v's subject NP has ***b*** as its value for attribute ***a***
3. Encode ***v***'s grammatical preferences as constraints in ILP

ILP formulation

For each test ***zp***, create an ILP program whose objective function is a linear combination of the ranker's probability, *P*(***op*** ,***c***), and the 4 probabilities encoding a verb's preferences, $P_{Num}(b)$, $P_{Gen}(b)$, $P_{Per}(b)$, and $P_{Ani}(b)$

$$\underset{op,c}{\operatorname{argmax}}\Big[\sum_{op\in PR}\sum_{c\in C} P(op,c)x(op,c) + \alpha\sum_{b\in V_{Num}} P_{Num}(b)y_{Num}(b) + \beta\sum_{b\in V_{Gen}} P_{Gen}(b)y_{Gen}(b) + \gamma\sum_{b\in V_{Per}} P_{Per}(b)y_{Per}(b) + \delta\sum_{b\in V_{Ani}} P_{Ani}(b)y_{Ani}(b)\Big] \quad (1)$$

subject to the following constraints:

$$x(op,c)\in\{0,1\}, \forall op\in PR, \forall c\in C \quad (2)$$

$$\sum_{op\in PR}\sum_{c\in C} x(op,c) = 1 \quad (3)$$

$$y_a(b)\in\{0,1\}, \forall a\in A, \forall b\in V_a \quad (4)$$

$$\sum_{b=1}^{|a|} y_a(b) = 1, \forall a\in A \quad (5)$$

$$y_{Ani}(animate) \geq y_{Gen}(masculine) + y_{Gen}(feminine) \quad (6)$$

(6) ensures that masculine or feminine pronouns are animate

$$\sum_{op_a=b}\sum_{c\in C} x(op,c) = y_a(b), \forall a\in A, \forall b\in V_a \quad (7)$$

(7) ensures consistency between the choice of ***op*** and the value $y_a(b)$ for each attribute ***a***

Evaluation

Corpus

- Training set: CoNLL-2012 shared task training data (1,391 documents; 12,111 **AZP**s)
- Test set: CoNLL-2012 shared task development data (172 documents; 1,713 **AZP**s)

Assumption

- Gold **AZP**s and gold syntactic parse trees are given

Evaluation metrics

- Recall (**R**), Precision (**P**), and F-measure (**F**)

Three supervised baseline systems

- Zhao and Ng (2007): Mention-pair model trained using syntactic and distance features
- Kong and Zhou (2010): Tree kernel-based method
- Chen and Ng (2013): Zhao and Ng's (2007) resolver augmented with two extensions

Results

	Baseline Systems									Our Approach					
	Kong and Zhou			Zhao and Ng			Chen and Ng			Ranking Model			ILP Method		
Source	R	P	F	R	P	F	R	P	F	R	P	F	R	P	F
Overall	44.9	44.9	44.9	41.5	41.5	41.5	47.7	47.7	47.7	45.9	46.4	46.1	**48.4**	**48.9**	**48.7**
NW	34.5	34.5	34.5	**40.5**	**40.5**	**40.5**	38.1	38.1	38.1	31.0	31.0	31.0	38.1	38.1	38.1
MZ	32.7	32.7	32.7	28.4	28.4	28.4	**34.6**	**34.6**	**34.6**	29.0	29.2	29.1	30.9	31.1	31.0
WB	45.4	45.4	45.4	40.1	40.1	40.1	46.1	46.1	46.1	45.4	45.4	45.4	**50.4**	**50.4**	**50.4**
BN	**51.0**	**51.0**	**51.0**	43.1	43.1	43.1	47.2	47.2	47.2	45.1	45.1	45.1	45.9	45.9	45.9
BC	43.5	43.5	43.5	44.7	44.7	44.7	52.7	52.7	52.7	50.2	50.7	50.4	**53.5**	**54.1**	**53.8**
TC	48.4	48.4	48.4	42.8	42.8	42.8	51.2	51.2	51.2	**53.7**	**56.1**	**54.9**	**53.7**	**56.1**	**54.9**

- Our ranking model beats Kong and Zhou's system and Zhao and Ng's system by 1.2% and 6.2% in F-score respectively, and only underperforms Chen and Ng's system by 1.6% in F-score
- Our ILP method beats all three baselines and achieves the best score on this dataset

Error analysis

- Incorrect choice of overt pronouns as gap fillers

(我) 想念(你)。 *pro1* 祝 ***pro2*** 平安。
(I) miss (**you**). *pro1* wish ***pro2*** are safe.

Our system mistakenly fills the gap of ***pro2*** with **op 我(I)** and further resolves ***pro2*** to candidate antecedent **我(I)**, while the correct antecedent should be **你(you)**.

- Incorrect resolution of ZPs with long-distance antecedents

(八里乡) 位于(((台北) 盆地) 西北端)。(行政区) 隶属于(台北县)， ***pro*** 为台北县廿九个乡镇市之一。
(**Bali Town**) is located in the (Northwest of ((Taipei) Basin)). (The administrative area) is affiliated with (Taipei County), ***pro*** is one of the 29 towns and cities.

Our system mistakenly resolves ***pro*** to its closest candidate antecedent **台北县(Taipei County)**, while the correct antecedent should be **八里乡(Bali Town)**.